Alagille Syndrome, Part Six: Kidney, Skeleton, Eyes and Face. Renal Dysplasia, Tubular Acidosis, Butterfly Vertebrae, Fragile Bones, Posterior Embryotoxon and the Facies.

Imagine a girl of four with Alagille syndrome, confirmed by a pathogenic variant in the gene JAG1. She still has her own liver. Her total bilirubin is about four milligrams per decilitre, and her height is two point four standard deviations below the average for her age. Last week she tripped on a rug at home and broke the shaft of her femur, the long bone of the thigh. Her blood tests show a serum bicarbonate of seventeen millimoles per litre, which is low, with a normal anion gap. The anion gap is a simple calculation from the sodium, chloride and bicarbonate that tells us whether an unmeasured acid has been added to the blood, as in lactic acidosis, or whether bicarbonate itself has been lost. A normal gap points to loss of bicarbonate, either through the gut or through the kidney. Her serum creatinine is zero point two milligrams per decilitre. An ophthalmologist has examined her eyes, found posterior embryotoxon and some pigment change in the peripheral retina, and written that no follow-up is needed. And her mother, looking at an old chest X-ray from her own childhood, has been told that one of her vertebrae has an unusual butterfly shape. The mother now wants to be the living liver donor.

The team has five questions. Is the fracture abuse, rickets, or a feature of Alagille bone, and what work-up does the child deserve? Is the low bicarbonate caused by diarrhoea from her ileal bile acid transporter inhibitor, the class of drug that blocks bile acid reabsorption in the last part of the small intestine and is now widely used for the itch of this disease, or is her kidney losing bicarbonate? Does a creatinine of zero point two really mean her kidneys are normal? Is the ophthalmologist right that her eyes need no follow-up? And what does the butterfly vertebra in her mother mean for the plan to use her as a donor?

The main idea of this chapter is simple. Outside the liver, heart and blood vessels, which were covered in earlier lessons, the kidney is the organ in Alagille syndrome that changes with time. It is involved in about four in every ten patients, with either of the two disease genes, and about half of the children whose kidney function has been measured after the age of two already have a reduced filtration rate. Liver transplantation then widens the gap. The spine, the eye and the face are different. They carry developmental signs that are present from birth and do not change much. These signs are valuable for making the diagnosis and for examining parents and siblings, but they are weaker in patients with the second gene, NOTCH2, and they rarely need treatment. Finally, the bones in Alagille syndrome break mainly because of cholestasis and poor growth rather than because of a primary bone defect. In practice, this means screening the kidney at diagnosis and before transplant, correcting acidosis, treating the bone through the liver, and using the developmental signs to examine relatives.

One ligand, five tissues, two kinds of problem

Alagille syndrome is caused in most patients by loss of one working copy of JAG1, which encodes a signalling protein called Jagged one. Jagged one sits on the surface of a cell and activates Notch receptors on the neighbouring cell, especially the receptor called Notch two. When one copy of JAG1 is lost, the signal runs at roughly half strength. In a small minority of patients the defect is in the receptor gene NOTCH2 itself. Development of many tissues depends on this signal, and the extrahepatic features of the syndrome are a map of where it matters.

In the kidney, mouse experiments show that Notch two is needed for cells of the developing nephron to take on a proximal identity, meaning to become podocytes, the filtering cells of the glomerulus, and the cells of the proximal tubule. In 2001, Brian McCright and colleagues reported in the journal Development that mice carrying one weakened copy of Notch2 together with one missing copy of Jag1 had kidneys about half the normal size, with fewer and abnormal glomeruli. This is the experimental basis for the renal dysplasia and tubular dysfunction seen in patients.

In the spine, each vertebral body forms from paired centres on the left and right. If the two halves fail to fuse in the midline, a cleft remains in the sagittal plane and the vertebra looks like the wings of a butterfly on a frontal X-ray. In Alagille syndrome these clefts are usually in the middle of the thoracic spine.

In the eye, posterior embryotoxon is a thickened and forward-displaced Schwalbe line. Schwalbe line is the ring where the inner lining of the cornea, called Descemet membrane, ends at the edge of the drainage angle of the eye. It is built from neural crest cells, the same migrating embryonic cells that build much of the face. The characteristic face of Alagille syndrome, with a broad forehead, deep-set eyes and a pointed chin, is also a product of neural crest development, and it changes with age.

Bone mass is different. In children, the clinical data suggest that low bone density is not mainly a primary Notch defect. In the multicentre North American study discussed below, the density deficit disappeared once the measurements were adjusted for the child's small body size, and the remaining variation tracked the level of bilirubin.

This gives us two kinds of problem. The spine, eye and face carry fixed developmental marks. We look for them once, we use them to support the diagnosis, and we use them to examine relatives. The kidney and the bone are dynamic. They worsen with cholestasis, malabsorption, calcineurin inhibitor drugs and transplantation, so they need regular surveillance and active treatment rather than a single look.

The kidney is the sign that does not fade in NOTCH2

The best modern picture of how often each feature occurs comes from the Global Alagille Alliance, usually called GALA, an international registry. In 2025, Sara Vandriel and colleagues published in Liver International a comparison of nine hundred and two patients with JAG1 variants and thirty-four patients with NOTCH2 variants. The characteristic face was recorded in about ninety percent of the JAG1 group but in only about fifty-eight percent of the NOTCH2 group. A structural heart anomaly on echocardiography was present in about ninety-two percent of JAG1 patients and sixty-four percent of NOTCH2 patients. Posterior embryotoxon was recorded in about fifty-three percent of JAG1 patients

and in only about nineteen percent of NOTCH2 patients. Butterfly vertebrae were found in about forty-five percent of JAG1 patients, but in only one of thirty NOTCH2 patients, which is about three percent. All of these differences were statistically significant. By contrast, renal anomalies were found in about thirty-nine percent of JAG1 patients and about thirty-five percent of NOTCH2 patients, and this difference was not significant. Vascular anomalies outside the heart also did not differ significantly, although numbers were small.

The message is that the kidney is the one feature that does not fade when the gene is NOTCH2. In NOTCH2 disease, the liver and the kidney carry the diagnosis, and the eye and the spine are often silent. A normal spine film or a normal slit-lamp examination therefore does not argue against NOTCH2-related Alagille syndrome. For comparison, the classic Philadelphia series of ninety-two patients, published by Kathleen Emerick and colleagues in Hepatology in 1999, reported the characteristic face in ninety-six percent, posterior embryotoxon in seventy-eight percent, butterfly vertebrae in fifty-one percent and renal disease in forty percent. The older numbers are higher for the eye and spine partly because those patients were identified clinically, which selects for people with the classic signs.

The kidney: four in ten are involved, and dysplasia leads

The key paper on the kidney was published in 2012 in the American Journal of Medical Genetics by Binita Kamath and colleagues in Philadelphia, under the title Renal anomalies in Alagille syndrome: a disease-defining feature. From a database of four hundred and sixty-six people with JAG1 variants, one hundred and eighty-seven had enough renal information to be evaluated. Seventy-three of them, which is thirty-nine percent, had renal involvement, and the authors considered this an underestimate because many patients had never been studied.

Among those seventy-three patients, the commonest finding by far was renal dysplasia, present in forty-three. Renal dysplasia means a kidney that did not develop normally, often small and echogenic on ultrasound, sometimes with cysts. It was generalised in twenty-eight and focal in eight. Renal tubular acidosis was found in seven patients, vesicoureteric reflux, meaning backflow of urine from the bladder toward the kidney, in six, and urinary tract obstruction in another six. Four patients had chronic renal failure and three had reached end-stage renal disease, one of whom needed a kidney transplant. Smaller numbers had bilateral renal artery stenosis, lipid deposits in the kidney, focal segmental glomerulosclerosis, which is scarring of parts of some glomeruli, or a duplicated collecting system.

The functional data were just as important. Among patients older than two years whose glomerular filtration rate could be estimated, eleven of twenty-two, which is half, had a filtration rate below ninety millilitres per minute per one point seven three square metres, the threshold below which kidney function is considered reduced. Among infants younger than two, four of seventeen were below the expected range for age. The authors proposed that renal involvement should be counted as the sixth defining criterion of Alagille syndrome, alongside the classical five, and recommended a targeted renal evaluation for

every patient, consisting of serum biochemistry, a renal ultrasound and a urinalysis.
A smaller Korean series, published in 2022 in Child Kidney Diseases by Joung and colleagues from Seoul, found renal involvement in ten of twenty-one children. Of the eight survivors with renal involvement, four, that is half, had progressed to chronic kidney disease stage three, meaning an estimated filtration rate between thirty and fifty-nine. The commonest renal lesion in Alagille syndrome is therefore structural and present from birth, and its commonest consequence is a filtration rate that falls quietly over the years.

Renal function before and after transplant

Two transplant studies show what happens next. The Studies of Pediatric Liver Transplantation registry, known as SPLIT, compared ninety-one children transplanted for Alagille syndrome with two hundred and thirty-six matched children transplanted for biliary atresia. The results were published by Kamath and colleagues in Liver Transplantation in 2012. After transplant, twenty-two percent of the Alagille recipients had a filtration rate below ninety, compared with eight percent of the biliary atresia recipients. One-year patient survival was also lower in Alagille syndrome, eighty-seven percent against ninety-six percent.
A single-centre study from Brussels, presented by Demaret and colleagues as an abstract in 2017, followed fifty children with Alagille syndrome. Twenty-eight underwent liver transplantation. Their mean estimated filtration rate after transplant was about ninety-four, compared with about one hundred and forty-three in children transplanted for other diseases. However, transplantation itself did not appear to lower filtration rate within the Alagille group, because children who were never transplanted had a similar mean of about eighty-eight. An abnormal renal ultrasound identified the children at highest risk: among those who were not transplanted, the mean filtration rate was about fifty-seven when the ultrasound was abnormal and about ninety-eight when it was normal.
Putting these together, the Alagille kidney enters transplantation already handicapped. The gap after transplant is mostly the native kidney disease, with calcineurin inhibitor toxicity added on top.
There is also a measurement trap. Serum creatinine is made by muscle, so a small child with poor muscle mass has a low creatinine even when filtration is reduced. In addition, high bilirubin causes negative interference with the older Jaffe method of measuring creatinine, so the reported value can be falsely low. A creatinine of zero point two in a jaundiced, small, sarcopenic child can therefore flatter the kidney. Cystatin C, a small protein made by all nucleated cells and filtered by the glomerulus, is much less dependent on muscle mass. The paediatric chronic kidney disease study group, known as CKiD, has published combined equations, called the CKiD under twenty-five equations, by Pierce and colleagues in Kidney International in 2021, that use creatinine, cystatin C or both. Before transplantation, a measured filtration rate, for example with iohexol clearance, is the most reliable value.

Acidosis: a normal-gap acidosis is a kidney question first

Renal tubular acidosis means that the tubules either fail to reclaim filtered bicarbonate, which is the proximal type, or fail to excrete acid, which is the distal type. In the Philadelphia series it was present in seven of the seventy-three patients with renal involvement, which is about one in ten. Both patients with tubular acidosis in the Seoul series had the proximal type. Dysplastic kidneys can also waste bicarbonate without a formal diagnosis of tubular acidosis.

The bedside approach is simple. A low bicarbonate with a normal anion gap and a high chloride means bicarbonate is being lost. The urine pH and the urine anion gap help separate renal loss from gut loss: when the gut is losing bicarbonate, a healthy kidney excretes more ammonium and the urine anion gap becomes negative, whereas in distal tubular acidosis the urine anion gap stays positive and the urine pH stays inappropriately high. Before blaming the tubule, the clinician should look for confounders. Diarrhoea from an ileal bile acid transporter inhibitor, the diuretic spironolactone, and dehydration can all lower the bicarbonate.

Why does it matter? A chronic acid load is buffered partly by bone, which releases calcium and carbonate. Acidosis also blunts the growth hormone and insulin-like growth factor axis, worsens linear growth, and promotes nephrocalcinosis, which is calcium deposition in the kidney. Treatment is simple: oral alkali, such as sodium bicarbonate or citrate, to keep the bicarbonate in the normal range. The 2012 chronic kidney disease guideline from KDIGO, Kidney Disease Improving Global Outcomes, suggests oral bicarbonate when the serum level is below twenty-two millimoles per litre. In a cholestatic child who is short and fracturing, a bicarbonate of seventeen is a treatable contributor to both problems, and tubular acidosis is one of the few kidney problems in Alagille syndrome with an easy and effective treatment. The bicarbonate should be rechecked after any change in drugs.

The spine: two in three have a vertebral anomaly

The most careful radiological study of the spine was published by Sanderson and colleagues from King's College Hospital in London in Pediatric Radiology in 2002. They reviewed the spine radiographs of fifty consecutive children with Alagille syndrome and thirty-one children with other forms of cholestasis. Sixty-six percent of the Alagille children had at least one vertebral anomaly, compared with under ten percent of the controls, a highly significant difference. Forty-eight percent of the Alagille children had two or more anomalies, with an average of two and a half per child. Of one hundred and twenty-six anomalies, one hundred and twenty-three were incomplete fusion of the front of the vertebral body, mostly between the sixth and ninth thoracic vertebrae. The most striking finding was that the original radiology reports were inaccurate in fifty-four percent of the Alagille children. In other words, the sign was there, but nobody had looked for it.

Butterfly vertebrae cause no symptoms, rarely progress, and almost never need treatment. Their value is diagnostic. In a cholestatic infant, a butterfly vertebra is strong support for Alagille syndrome, but only if someone deliberately looks for it. The practical step is to request a frontal view of the thoracic spine and tell the radiologist what to look for. Remember also the gene difference: butterfly vertebrae are found in nearly half of JAG1 patients and almost never in NOTCH2 patients.

Bone fragility: low-trauma fractures of the legs

Children with Alagille syndrome break bones easily. Bales and colleagues in Philadelphia, writing in the Journal of Pediatric Gastroenterology and Nutrition in 2010, surveyed families in their research database. Twelve of forty-two children, which is twenty-eight percent, had a total of twenty-seven fractures. The average age at fracture was five years. Eighty-four percent of the fractures happened with little or no trauma, and seventy percent involved the long bones of the legs. The rate of femur fracture was about one hundred and twenty-eight per ten thousand person-years. The authors acknowledged that the cohort was small and that families of children with fractures may have been more likely to respond.
Why are the bones weak? The Childhood Liver Disease Research Network, known as ChiLDReN, measured bone density by dual-energy X-ray absorptiometry, or DXA, in children with several chronic liver diseases. The study was published by Kathleen Loomes and colleagues in Hepatology in 2019. Children with Alagille syndrome had low bone mineral density and content compared with children with other diseases, but this deficit disappeared once the values were adjusted for height and weight. In other words, most of the deficit reflected small bones in a small child. Within the Alagille group, however, the size-adjusted values fell as bilirubin and serum bile acids rose, and they were lower in children who had already fractured.
There is an interesting contrast in the genetics. Gain-of-function variants in NOTCH2, which make the receptor overactive, cause a completely different disease called Hajdu-Cheney syndrome, with severe osteoporosis and dissolution of the tips of the fingers, as reported by two groups in Nature Genetics in 2011. The same gene can therefore affect bone in opposite ways depending on the direction of the defect.
The clinical conclusion is that the Alagille skeleton is small and cholestatic rather than intrinsically brittle. Fracture risk follows bilirubin, vitamin D, acidosis and growth, which are all things the hepatologist can change. A low-trauma femoral fracture in a jaundiced child with Alagille syndrome is plausibly pathological, but it still needs a complete and documented safeguarding assessment, because children with chronic illness can also be abused.

Protecting bone: treat the bone through the liver

Each driver of poor bone health has a matching action. Vitamin D is a fat-soluble vitamin, so it is poorly absorbed in cholestasis; the twenty-five hydroxy vitamin D level should be checked at each visit, water-miscible preparations and higher doses are often needed, and the level should be rechecked after each change in dose. Cholestasis itself drives bone loss, so reducing it helps bone: an ileal bile acid transporter inhibitor, which the 2024 European Association for the Study of the Liver guideline on genetic cholestatic liver diseases strongly recommends offering for cholestatic pruritus, surgical biliary diversion, or transplantation when indicated. Growth and calories matter, because most of the density deficit is small bones in a small child; energy-dense feeds with medium-chain triglycerides, fats that can be absorbed without bile, are part of bone care. Acidosis should be corrected with alkali. Vitamin K is needed not only for clotting but also for proteins in the bone matrix, and supplementation guided by the international normalised ratio covers both.

Finally, a raw DXA Z-score overstates the deficit in a short child, so the result should be adjusted for height-for-age and compared with the child's own previous scans. The rule is to correct vitamin D, acid, calories and cholestasis before considering bisphosphonates, because most of the bone deficit follows them.

The eye: embryotoxon is only the most visible finding

The classic prospective eye study was published by Melanie Hingorani, Ken Nischal and colleagues from Great Ormond Street Hospital in Ophthalmology in 1999. They examined twenty-two children with Alagille syndrome and twenty-three of their parents. Posterior embryotoxon was found in ninety-five percent of the children. Optic disc anomalies were found in seventy-six percent, diffuse hypopigmentation of the back of the eye in fifty-seven percent, iris abnormalities in forty-five percent, and speckling of the retinal pigment epithelium, the pigmented cell layer beneath the retina, in thirty-three percent. At least one of these signs was found in thirty-six percent of the parents. Vitamin A and E levels were normal, which argued against vitamin deficiency as the cause, and visual acuity was not significantly affected. Two years earlier, Nischal and colleagues had described optic disc drusen, small deposits at the optic nerve head, on ocular ultrasound as a new sign of the syndrome.

The difference between ninety-five percent in this slit-lamp series and about fifty-three percent in the GALA registry reflects how carefully the eye is examined. A slit lamp finds embryotoxon; a dilated examination of the back of the eye finds the optic disc and retinal signs, which are more specific to Alagille syndrome.

The retina: no serious functional significance is not always true

Newer retinal imaging studies have challenged the view that the eye findings are harmless. In 2021, da Palma and colleagues, in a multicentre study published in Investigative Ophthalmology and Visual Science, imaged twenty-three patients with Alagille syndrome. Ninety-six percent had peripheral chorioretinal changes, which were more common than embryotoxon. The median visual acuity was twenty over twenty-five, which is near normal, but the range extended down to the ability only to see hand movements, and ninety-one percent had ocular symptoms, most often loss of peripheral vision. Other reports have described recurrent bleeding within the retina in three patients, suggesting the same fragile small vessels seen in the brain, and a case series has described visually significant atrophy of the macula, the central part of the retina, in a minority.

The honest position is that the older prospective series found little loss of function, while the newer imaging series come from retina clinics and tend to select patients with symptoms. The truth for an unselected child probably lies in between. A baseline dilated eye examination at diagnosis and periodic review in adolescence cost little and will catch the minority who lose peripheral field or central vision. So for our patient, the ophthalmologist's note of no follow-up deserves a second look.

The face: specific in expert eyes, unreliable outside them

The facial features of Alagille syndrome are a broad forehead, deep-set eyes, mildly increased distance between the eyes, a straight nose with a bulbous tip, and a pointed chin, which together give a triangular shape. For years it was debated whether this was a specific syndromic face or simply the face of any child with chronic cholestasis. In 2002, Kamath and colleagues asked clinical dysmorphologists, who were blinded to the diagnosis, to classify photographs of eighteen patients with Alagille syndrome or other forms of congenital intrahepatic cholestasis. The examiners were correct seventy-nine percent of the time, with a sensitivity of seventy-six percent and a specificity of eighty-two percent. Accuracy was lowest for adult faces, where the chin becomes more prominent and the triangle softens. The authors concluded that the face is specific to the condition.

A study published in 2012 by Lin and colleagues tested the face in a different population, Vietnamese children with Alagille syndrome. When thirty-seven North American dysmorphologists judged twenty photographs, overall agreement on whether a child had the syndrome was only about fifty-two percent, and only four faces achieved agreement above seventy-five percent. A Vietnamese paediatric gastroenterologist recognised the facial phenotype in sixty-one percent of the JAG1-positive children, whereas a visiting dysmorphologist recognised it in only twenty-four percent. In the GALA cohort, the face was recorded in about ninety percent of JAG1 patients and fifty-eight percent of NOTCH2 patients.

The face is real and specific in experienced hands, but it depends on age, ancestry and the examiner. It should be photographed and compared with the parents, and it should never outweigh or replace genetic testing.

Weighing the signs at the bedside

It helps to set each sign against how often it appears in people without the syndrome. The characteristic face is seen in about ninety percent of JAG1 patients and fifty-eight percent of NOTCH2 patients, and experts are about eighty-two percent specific when distinguishing it from other cholestasis. Posterior embryotoxon is found in between about half and nearly all patients depending on how carefully it is sought, and in only about one in five NOTCH2 patients, but it is also commonly quoted as present in eight to fifteen percent of the general population, so it is supportive only, and is most useful when looking at the parents. Butterfly vertebrae are found in about forty-five percent of JAG1 patients and three percent of NOTCH2 patients, and any vertebral anomaly was present in under ten percent of cholestatic controls in the London study, so a butterfly vertebra is strong support if it is deliberately sought. Renal anomalies are found in about thirty-nine percent of JAG1 patients and thirty-five percent of NOTCH2 patients; they are not specific on their own, but they persist in NOTCH2 disease and have been proposed as a sixth criterion. Retinal and optic disc changes are frequent in the syndrome and uncommon in other cholestasis, but they are underused because they need a dilated examination.

The rule is that signs raise or lower the probability before testing; genetic testing of JAG1 and NOTCH2 makes the diagnosis; and a missing sign never excludes Alagille syndrome.

A surveillance plan and the transplant

No graded guideline sets out surveillance for these organs. The only published recommendation is the Philadelphia group's call for a renal ultrasound, urinalysis and blood biochemistry in every patient. The rest of the schedule offered here is expert practice built from the cohort data. At diagnosis, it is reasonable to obtain a renal ultrasound, a urinalysis, a blood pressure, a creatinine with cystatin C, and a bicarbonate; a spine radiograph read by someone who knows what to look for; and both a slit-lamp and a dilated fundus examination. At every visit, growth, the vitamin D level, the international normalised ratio, vitamins A and E, the bicarbonate, a urinalysis and blood pressure in both arms should be checked. Each year, an estimated filtration rate based on cystatin C or a combined equation, and a urine protein measurement, are sensible, with a height-adjusted DXA scan if the child has fractured or has had long-standing cholestasis. Before transplant, the child needs a measured filtration rate, imaging of the kidneys including the renal arteries, a nephrology review, and a plan for kidney-sparing immunosuppression. The family also matters: a parent offering to donate should have a slit-lamp examination, a spine film, an echocardiogram and a renal ultrasound, plus targeted testing for the family variant.

The transplant decision deserves emphasis. Because creatinine-based estimates overestimate kidney function in cholestatic, sarcopenic children, cystatin C or a measured filtration rate should be obtained before listing. If the filtration rate is already reduced, the team should plan lower calcineurin inhibitor exposure from the start, using mycophenolate or a drug of the mTOR inhibitor class, such as everolimus. Combined liver and kidney transplantation is uncommon; the Philadelphia series reported one kidney transplant for end-stage disease and two more after liver transplantation. In Alagille syndrome, transplant assessment is a three-organ assessment, of heart, vessels and kidney, and the kidney is the one most often under-measured. A parent donor needs the same extrahepatic screen, because a mildly affected parent may have a dysplastic kidney or a liver with subclinical disease.

What the guidelines say

The 2017 joint guideline of the North American and European societies for paediatric gastroenterology, hepatology and nutrition, known as NASPGHAN and ESPGHAN, on the evaluation of cholestatic jaundice in infants, includes a spine radiograph for butterfly vertebrae and an echocardiogram among its second-tier tests, lists ophthalmology consultation, and includes the characteristic face among the clinical criteria. The 2024 European Association for the Study of the Liver guideline on genetic cholestatic liver diseases lists renal dysplasia, vertebral anomalies, posterior embryotoxon and facial dysmorphism as features, but gives no recommendation on how to evaluate or follow them. The 2014 American transplant evaluation guideline from AASLD, AST and NASPGHAN asks for an assessment of renal function before transplant but nothing specific to Alagille syndrome.

So the documents agree on the spine film. They differ from expert evidence on the eye, where the literature argues for a dilated fundus examination as well as a slit lamp, and on the face, where the Vietnamese data show that facial features are unreliable across ancestries. The material gap is the kidney. The guidelines list it as a feature but never ask

anyone to look at it, although it is involved in four in ten patients and its function falls before transplant. There is also no guidance on how to measure filtration rate or how to protect bone.

Traps and pearls

Several traps recur. Reading a creatinine of zero point two in a jaundiced, small child as normal renal function. Blaming a low bicarbonate on bile acid transporter inhibitor diarrhoea without checking the urine. Accepting a report of no vertebral anomaly from a chest film that nobody examined for clefts. Using an absent butterfly vertebra or embryotoxon to argue against NOTCH2 disease. Treating embryotoxon as the whole eye examination and never dilating the pupil. Calling a low-trauma femoral fracture abuse without considering cholestatic bone, or calling it bone disease without a safeguarding assessment. Reading an unadjusted DXA Z-score in a very short child and starting bisphosphonates. Listing for transplant without a measured filtration rate or a kidney-sparing plan. And accepting a parent donor without a slit lamp, a spine film, an echocardiogram and a renal ultrasound.
The numbers worth remembering are these. Renal anomalies affect about thirty-nine percent of JAG1 patients and thirty-five percent of NOTCH2 patients, the only sign with no gene gap. Renal dysplasia is the leading lesion, in forty-three of seventy-three patients with renal involvement, and tubular acidosis occurs in seven of seventy-three. Half of evaluable JAG1 children older than two have a filtration rate below ninety. After liver transplant, twenty-two percent of Alagille recipients have a filtration rate below ninety, against eight percent of biliary atresia recipients. Vertebral anomalies are seen in sixty-six percent of Alagille children against under ten percent of cholestatic controls, and more than half of the original reports were inaccurate. Butterfly vertebrae occur in about forty-five percent of JAG1 patients but three percent of NOTCH2 patients. Fractures occurred in twenty-eight percent of a family-survey cohort, eighty-four percent with little or no trauma, at an average age of five. The bone density deficit disappears after adjustment for size and tracks bilirubin. Embryotoxon was found in ninety-five percent of a slit-lamp series and about fifty-three percent of the registry, and peripheral retinal change in ninety-six percent of an imaging series. Expert recognition of the face has a sensitivity of about seventy-six percent and a specificity of about eighty-two percent, and does much worse across ancestries.

What to do differently in clinic

Return to the four-year-old girl. Check her urine pH and urine anion gap, and if her kidney is losing bicarbonate, start alkali and keep her bicarbonate in the normal range. Measure cystatin C rather than trusting the creatinine of zero point two, and arrange a renal ultrasound if she has not had one. Replace her vitamin D, review her response to the bile acid transporter inhibitor and her calorie intake, and document a full safeguarding assessment for the fracture, while recognising that cholestatic bone is a likely explanation. Book a dilated eye examination rather than accepting a single slit-lamp look. And before her mother is considered as a living donor, screen the mother's eyes, spine, heart and

kidneys, and test her for the family variant, because a butterfly vertebra in a parent may be the first sign that the parent carries the same disease.

The broader lesson for clinic is to treat the extrahepatic features of Alagille syndrome as two groups. The spine, eye and face are developmental signs: look for them carefully once, use them to support the diagnosis and to examine relatives, and remember that they fade in NOTCH2 disease. The kidney and bone are living problems: measure the kidney properly at diagnosis, every year and before transplant, correct acidosis, and protect bone by controlling cholestasis, vitamin D and growth.